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EXTENDED ABSTRACT

Towards Sensor-Agnostic Precipitation Prediction using Passive Microwave Satellites

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1. Introduction and objectives

Reliable, low-latency precipitation estimation is increasingly vital as climate change exacerbates water scarcity and intensifies weather variability, leading to more frequent unforeseen disasters such as floods. This is especially true for low-income regions, where ground-based precipitation-measuring instruments are entirely absent, as well as in remote and inaccessible areas such as the open ocean which strongly influences terrestrial weather. In such contexts, satellite-based observations provide the only viable means of achieving global and accurate precipitation coverage. The Global Precipitation Measurement (GPM) constellation [4] addresses this need through a fleet of satellites that continuously observe the Earth and measure microwave brightness temperatures. Amongst them, the GPM Core Observatory (GPM-Core) satellite plays a unique role by also providing very precise raining rate estimates using a dual-frequency precipitation radar. However this equipment only orbits over a small region of the Earth every few days, and this hinders its maximum potential.

Some works such as DRAIN [7] have sought to extend the effective swath by using deep neural networks to estimate raining rates from brightness temperature observations. However this methods only works on the GPM-Core satellite, and is not applicable to other satellites such as the F18[1] — despite having similar acquisition frequencies — due to significant differences in orbit characteristics and data distributions. To address this issue, the method [5] transformed F18 observation into GPM ones, by employing generative models such as CycleGAN [8]. Nevertheless, these models are notoriously difficult to train due to instability issues, which limits scalability to satellites beyond F18. In this work, we seek to propose a novel approach for unsupervised domain adaptation as a promising alternative that could overcome these limitations.

2. Methods and data

Our proposed approach relies on deep generative models, specifically conditional flow matching[3], to perform unsupervised domain adaptation from F18 satellite observations to the GPM-Core observation domain. The choice of flow matching is motivated by its strong performance relative to diffusion models [2], as well as its greater training stability compared to CycleGAN [8]. The generative model is trained using GMI and F18 observations from 2018, focusing on precipitation-sensitive channels near

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37 and 89 GHz with dual polarization. During inference, the flow matching model modifies the content and the style of an F18 observation so that its distribution more closely aligns with that of GPM observations. This is achieved in two steps: first, the image is noised according to its source label (F18); second, it is denoised using the target label (GPM), following a strategy similar to that proposed in [6].

3. Preliminary results and/or expected outcomes

Qualitative assessments indicate that the proposed method enhances the spatial resolution and structural consistency of the adapted images. This results in a performance increase on raining rates retrieved by applying DRAIN on the adapted F18 images compared to the raining rates obtained using the non adapted images. This is illustrated in Figure 1, where the retrieved rain structure and intensity more closely match those derived from native GMI observations.

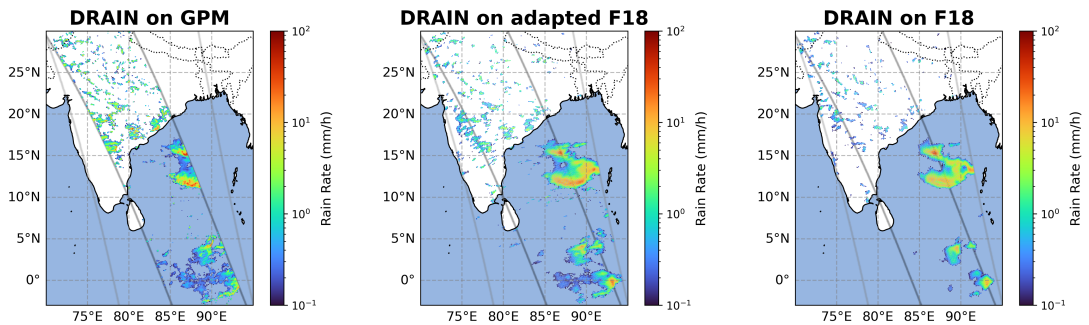

Figure 1. *Comparison of raining rates retrieved using DRAIN from (i) GMI observations (left), (ii) adapted F18 observations obtained via our conditional flow matching approach (middle), and (iii) the original, non-adapted F18 observations (right). Raining rates retrieved from the adapted F18 images exhibit enhanced structural detail and more accurate intensity estimates than from original F18 images. The observation date is 31/08/2019.*

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